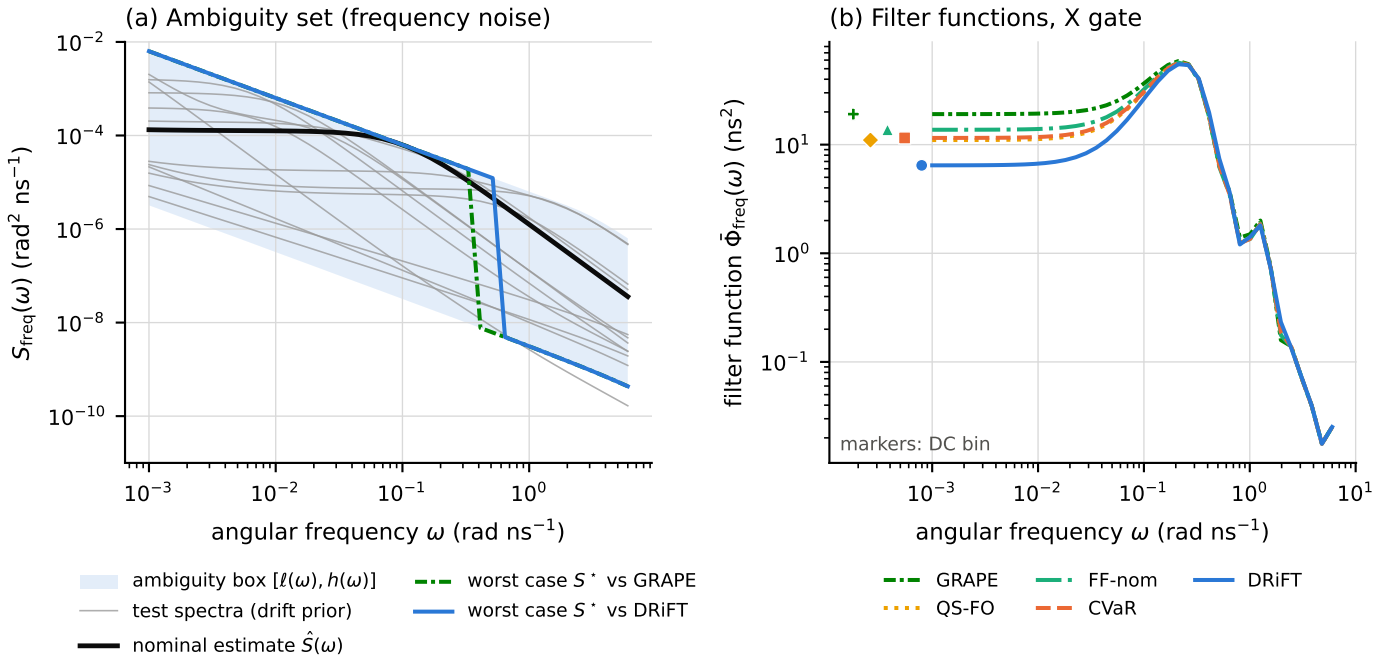
(a) Ambiguity set (frequency noise)
$S_{\mathrm{freq}}(\omega)$ (rad$^2$ ns$^{-1}$)
$10^{-2}$
$10^{-4}$
$10^{-6}$
$10^{-8}$
$10^{-10}$
$10^{-3}$
$10^{-2}$
$10^{-1}$
$10^{0}$
angular frequency $\omega$ (rad ns$^{-1}$)
ambiguity box $[\ell(\omega), h(\omega)]$
test spectra (drift prior)
nominal estimate $\hat{S}(\omega)$
worst case $S^\star$ vs GRAPE
worst case $S^\star$ vs DRiFT
(b) Filter functions, X gate
filter function $\bar{\Phi}_{\mathrm{freq}}(\omega)$ (ns$^2$)
$10^{1}$
$10^{0}$
$10^{-1}$
markers: DC bin
$10^{-3}$
$10^{-2}$
$10^{-1}$
$10^{0}$
$10^{1}$
angular frequency $\omega$ (rad ns$^{-1}$)
GRAPE
FF-nom
DRiFT
QS-FO
CVaR